b. Working principle

When the supply is switched on, full voltage is applied across main and starting electrodes. This voltage fills the gap between the electrodes and discharge through argon gas. As the lamp warms up, mercury is vaporized, which increases the vapour pressure. After 5 minutes, the lamp gives full light. It gives a greenish blue colour light. This lamp is always kept in vertical. Otherwise the inner glass tube may break due to excess heat.

c. Advantages

1. Mercury vapour lamps are more energy efficient than incandescent lamps.
2. It has high luminous efficacies of 35 to 65 lumens / watt.
3. It is durable. (in the range of 24,000 hours)
4. It has a high intensity.
5. It gives clear white light output which has made them ideal for outdoor use

d. Applications

1. Mercury vapour lamps are used in lighting applications.
2. It is used in streets and parking places.
3. It is used for landscape lighting.
4. It is used in factories.
5. It is used in gymnasiums.

2.7 Fluorescent lamp and compact fluorescent lamp

2.7.1 Fluorescent lamp

It is a low pressure mercury vapour lamp. It consists of a glass tube 25 mm in diameter and 0.6 m, 1.2 m and 1.5 m in length. The inner portion of the tube is coated with phosphorous. The tube contains argon gas at low pressure and a drop of mercury is added. The choke, two filaments and the starter are connected in series as shown in the figure 2.8.

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a. Working principle

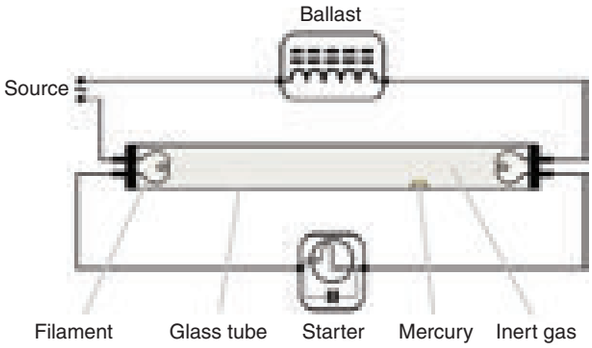


Fig. 2.8 Connection diagram a fluorescent lamp

Fluorescent lamp function based on the principle of current passing through air medium. In the air medium the resistance falls down heavily. When the current passes to the lamp, circuit is closed through choke, filament and starter. So the 230V supply voltage is applied between the starter terminals. Due to this voltage, current starts flowing through the inert gas in the starter. Now the bimetallic strips raised touch each other as the temperature of inert gas has increased. Once the circuit is completed through the bimetallic strips and filaments in the tube light, the bulb will start glowing.

The inert gas in the starter cools down and the bimetallic strip opens again. Therefore the current through the choke decreases and hence the magnetic flux decreases. The decreasing (or alternating) flux is cut by the choke winding which causes self induced EMF in the choke coil.

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